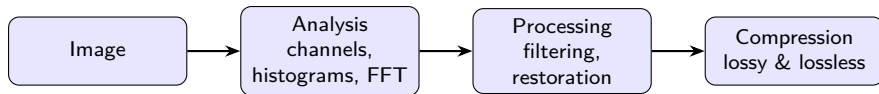


An Image Processing Workspace

Signals, Transforms, and Compression: Concepts and Algorithms

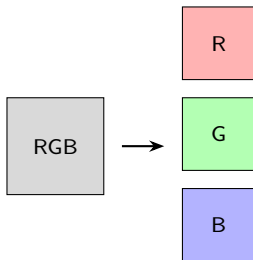
The Big Picture

- An image is a 2D signal: analyze it, transform it, compact it.
- Every tool here is one of those three moves.



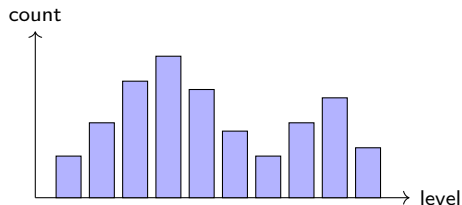
Channel Separation

- RGB = three stacked signals $R(x, y)$, $G(x, y)$, $B(x, y)$.
- BGR vs. RGB load order is a classic bug, caught by viewing channels.



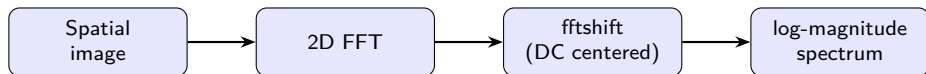
Histograms: Statistics of a Signal

- Histogram = empirical distribution of pixel intensities.
- Luminance blends channels *per pixel first*: $Y = 0.299R + 0.587G + 0.114B$.



Frequency Content: The 2D FFT

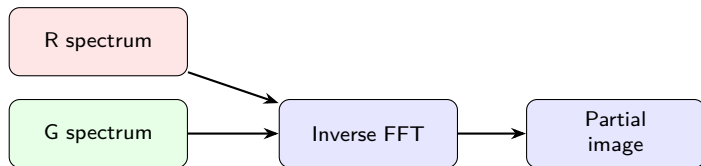
- Image $\rightarrow$ sum of 2D sinusoids of varying spatial frequency.
- `fftshift` centers DC; log-magnitude makes the spectrum viewable.



Parseval: energy in space = energy in frequency

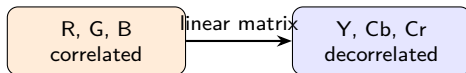
Partial Reconstruction

- Inverse FFT is linear: rebuild from a subset of channel spectra.
- *Phase*, not magnitude, carries most of the structure.



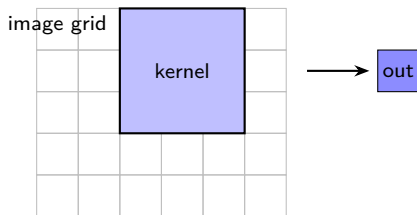
Color Space: RGB vs. YCbCr

- Linear change of basis: decorrelates brightness from color.
- Y carries detail; C_b, C_r are smooth $\Rightarrow$ energy compaction.



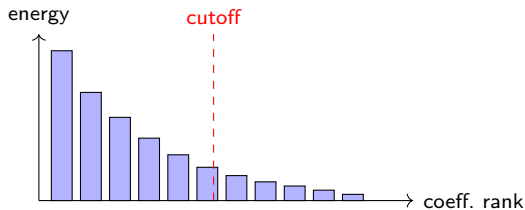
Spatial Filtering: Convolution

- A kernel slides over the image; each pixel becomes a weighted local sum.
- Low-pass = blur (box, Gaussian); high-pass = sharpen (Laplacian, unsharp).



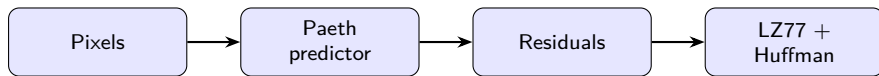
Lossy Compression: Fourier Energy Pruning

- Rank frequency coefficients by energy; keep a target energy %.
- Real images: conjugate-symmetric spectrum $\Rightarrow$ store half.



Lossless Compression: PNG-Style Codec

- Predict each pixel from neighbors (Paeth); store the residual.
- Residuals compress far better under LZ77 + Huffman coding.



CRC-32 checks integrity of every chunk

Lossless Compression: Integer Wavelet Lifting

- Multiresolution: split into coarse *approximation* + *detail*.
- Lifting uses only integer add/subtract $\Rightarrow$ exactly reversible.

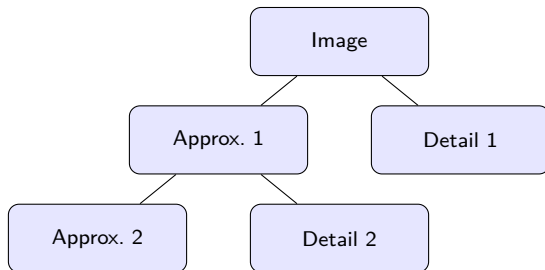
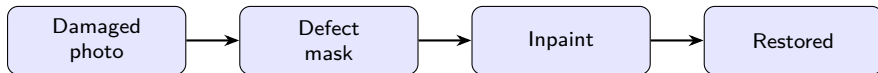


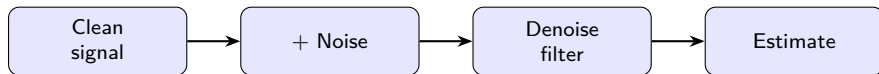
Photo Restoration

- Morphology (top-hat/black-hat) detects small bright/dark defects.
- Inpainting fills the mask; notch filtering removes periodic bands.



Noise: Models and Denoising

- Gaussian noise: additive per-pixel; salt-and-pepper: outlier pixels.
- Median (nonlinear) beats blur (linear) on outliers; NL-means uses self-similarity.



Recurring Threads

